

- Description

This module manages a smart parking lot vacancy monitoring system, ensuring that:

1. The system continuously monitors the number of available parking spaces.
2. If the number of available spaces falls below a predefined minimum threshold, an alert is triggered to indicate that the parking lot is nearly full.
3. If the parking lot is completely full (zero spaces available), a "Full" sign is activated, preventing further entry.
4. If sufficient spaces are available, no action is required.

This system helps efficiently manage parking availability and prevent congestion.

- Specification

module SmartParkingLotMonitoring

var availableSpaces, alertThreshold: integer

var fullSignActive: boolean

process updateAvailableSpaces(newAvailableSpaces: integer) returns
(availableSpaces: integer, fullSignActive: boolean)

functional scenario form:

availableSpaces = 0 && fullSignActive = true, output "ALERT: Parking lot full!
No more vehicles allowed.", Full sign activated.

||

availableSpaces < alertThreshold && output "WARNING: Parking lot almost
full. Limited spaces available.", Parking lot warning triggered.

||

availableSpaces >= alertThreshold && fullSignActive = false, output "Parking
lot has sufficient space.", Parking lot operating normally.

end process

end module

- Java Code

```
package code;
public class c110_SmartParkingLot {
    private int totalSpaces;
    private int availableSpaces;
    private int alertThreshold;
    private boolean fullSignActive;
    public c110_SmartParkingLot(int totalSpaces, int alertThreshold, int
initialAvailableSpaces) {
        this.totalSpaces = totalSpaces;
        this.alertThreshold = alertThreshold;
        this.availableSpaces = initialAvailableSpaces;
        this.fullSignActive = false;
    }
    public void updateAvailableSpaces(int newAvailableSpaces) {
```

```

        availableSpaces = newAvailableSpaces;
        adjustParkingStatus();
        System.out.printf("Available Spaces: %d, Full Sign: %s\n",
            availableSpaces, fullSignActive ? "ON" : "OFF");
        if (fullSignActive) {
            System.out.println("ALERT: Parking lot full! No more vehicles allowed.");
        } else if (availableSpaces < alertThreshold) {
            System.out.println("WARNING: Parking lot almost full. Limited spaces available.");
        } else {
            System.out.println("Parking lot has sufficient space.");
        }
    }
    private void adjustParkingStatus() {
        if (availableSpaces == 0) {
            fullSignActive = true;
        } else {
            fullSignActive = false;
        }
    }
    public int getAvailableSpaces() {
        return availableSpaces;
    }
    public static void main(String[] args) {
        c110_SmartParkingLot parkingLot = new c110_SmartParkingLot(100, 10, 50);

        // testcase-VT:
        // parkingLot.updateAvailableSpaces(8);
        // parkingLot.updateAvailableSpaces(0);
        // parkingLot.updateAvailableSpaces(15);

        // testcase-FT:
        // parkingLot.updateAvailableSpaces(1280);
        // parkingLot.updateAvailableSpaces(129);
        // parkingLot.updateAvailableSpaces(5);

        // testcase-Z3:
        // parkingLot.updateAvailableSpaces(8);
        // parkingLot.updateAvailableSpaces(0);
        // parkingLot.updateAvailableSpaces(15);

        // testcase-UVT:
        parkingLot.updateAvailableSpaces(8);
    }

```

```

        parkingLot.updateAvailableSpaces(0);
        parkingLot.updateAvailableSpaces(15);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c110_SmartParkingLot.java	100.0 %	8	0	8
▼ c110_SmartParkingLot	100.0 %	8	0	8
● updateAvailableSpaces(int)	100.0 %	6	0	6
■ adjustParkingStatus()	100.0 %	2	0	2
● c110_SmartParkingLot(int, int, in		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,938	22,027
▼ src	0.4 %	89	21,938	22,027
▼ code	0.4 %	89	21,938	22,027
▼ c110_SmartParkingLot.java	96.7 %	89	3	92
▼ c110_SmartParkingLot	96.7 %	89	3	92
● updateAvailableSpaces(int)	100.0 %	46	0	46
■ adjustParkingStatus()	100.0 %	11	0	11
● c110_SmartParkingLot(int, int, in	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

4. Fuzz testing branch coverage: 62.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c110_SmartParkingLot.java	62.5 %	5	3	8
▼ c110_SmartParkingLot	62.5 %	5	3	8
● updateAvailableSpaces(int)	66.7 %	4	2	6
■ adjustParkingStatus()	50.0 %	1	1	2
● c110_SmartParkingLot(int, int, in		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 85.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	79	21,948	22,027
▼ src	0.4 %	79	21,948	22,027
▼ code	0.4 %	79	21,948	22,027
▼ c110_SmartParkingLot.java	85.9 %	79	13	92
▼ c110_SmartParkingLot	85.9 %	79	13	92
● updateAvailableSpaces(int)	87.0 %	40	6	46
■ adjustParkingStatus()	63.6 %	7	4	11
● c110_SmartParkingLot(int, int, in	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c110_SmartParkingLot.java	100.0 %	8	0	8
▼ c110_SmartParkingLot	100.0 %	8	0	8
● updateAvailableSpaces(int)	100.0 %	6	0	6
■ adjustParkingStatus()	100.0 %	2	0	2
● c110_SmartParkingLot(int, int, in		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,938	22,027
▼ src	0.4 %	89	21,938	22,027
▼ code	0.4 %	89	21,938	22,027
▼ c110_SmartParkingLot.java	96.7 %	89	3	92
▼ c110_SmartParkingLot	96.7 %	89	3	92
● updateAvailableSpaces(int)	100.0 %	46	0	46
■ adjustParkingStatus()	100.0 %	11	0	11
● c110_SmartParkingLot(int, int, in	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c110_SmartParkingLot.java	100.0 %	8	0	8
▼ c110_SmartParkingLot	100.0 %	8	0	8
● updateAvailableSpaces(int)	100.0 %	6	0	6
■ adjustParkingStatus()	100.0 %	2	0	2
● c110_SmartParkingLot(int, int, in		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,938	22,027
▼ src	0.4 %	89	21,938	22,027
▼ code	0.4 %	89	21,938	22,027
▼ c110_SmartParkingLot.java	96.7 %	89	3	92
▼ c110_SmartParkingLot	96.7 %	89	3	92
● updateAvailableSpaces(int)	100.0 %	46	0	46
■ adjustParkingStatus()	100.0 %	11	0	11
● c110_SmartParkingLot(int, int, in	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17